Cosmic Muons

Author: Mark Mathison
Lab Partner: Yoon Gyu Lee

November 12, 2024

Abstract

This experiment utilizes a scintillator telescope to detect cosmic muon rays. The rate of cosmic muons observed at the Earth's surface exhibits a cosine-squared relationship with the zenith angle. In this work, the angular dependence of the rate of cosmic muon rays was observed in good agreement with a cosine-squared law. In addition, instrumental techniques of the scintillator telescope were explored, including determining optimal discriminator thresholds, establishing nominal counting rates of component detectors, and performing a precision calibration of a TAC module which could be used in the future to perform measurements of muon decays.

1 Background Introduction

1.1 Motivation

The Earth is being showered in cosmic radiation at all times. These cosmic rays can be categorized into primary and secondary particles, where primary particles are direct products of astrophysical sources, and secondary particles are produced upon primary particles interacting with the Earth's atmosphere. Examples of primaries include electrons, protons, helium, and other light nuclei synthesized in stars via fusion, while secondaries include light nuclei not synthesized in stars, such as lithium, beryllium, and boron, as well as positrons, antiprotons, and pions. If charged pions decay in the atmosphere around 15km above the Earth's surface, they can produce cosmic muons [1].

Muons are unstable particles with an average lifespan of $2.2\mu s$. Classically speaking, a muon traveling a the speed of light would then decay after only 660m, but when considering relativistic corrections for time dilation, they travel an average distance of 33km before decaying [1]. This allows them to cover the distance from their point of origin at the top of the Earth's atmosphere to the Earth's surface. The muon decay is also of interest as it is one of the few observable processes which is dictated only by the Weak force. As such, muon decays can serve as a probe into the strength of the Weak force.

The observed rate of cosmic muons at the Earth's surface exhibits a notable dependence on the muon's angle of incidence. This dependence comes from a combination of two factors: the length of the muon's path through the atmosphere, and the direction of the incident pion which produced the muon. These two effects together cause the rate of muons at the Earth's surface to be approximately proportional to the cosine squared of the zenith angle.

1.2 Summary of Experiment

This experiment utilizes a scintillator telescope to measure the rates of cosmic muons as a function of zenith angle. Each scintillator paddle has two PMTs whose signals flow into discriminators and a series of coincidence units. First, one must determine a nominal rate of signals for the scintillator telescope. To do this, the threshold voltages of the discriminators were set to filter out noise. Then, the signal rates were measured for each individual PMT, followed by measuring the signal rates of the two PMTs on each scintillator paddle in coincidence. Finally, the event rates for 4-fold coincidences between all the PMTs were recorded. After establishing the nominal rates of the scintillator telescope, the angular dependence of muon rates was investigated by recording 4-fold coincidences at varying zenith angles. Lastly, a calibration was performed for the TAC module using 4-fold coincidences with a duplicated signal with a manually adjusted time offset. This process is necessary for using a scintillator telescope to measure the muon lifetime, but due to the technical limitations of the telescope used, this measurement was not performed.

2 Theory

The observed rate of cosmic muons at the Earth's surface is primarily determined by the muon's path through the Earth's atmosphere. Due to the effects of the special theory of

relativity, these particles travel on average around 33km before decaying, despite having a mean lifetime of only $2.2\mu s$. The distance muons have to travel to reach an observer on the Earth's surface depends on the angle between the muon's path and the zenith for the observer. Cosmic muons are created by pions decaying at an altitude of around 15km, so when the zenith angle is near 0°, the muon only needs to travel around 15km to reach the observer. Meanwhile, if the zenith angle is near 90°, the muon has to travel around 400km to reach the observer, which is only achievable if the muons are exceptionally high-energy [1]. Thus, muons are more likely to decay before reaching the observer at larger zenith angles, reducing the observed rates. Figure 1 illustrates this point.

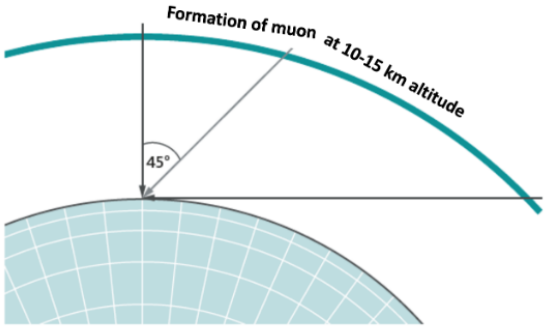


Figure 1: Diagram showing a muon's path from its origin in the Earth's atmosphere to an observer on the Earth's surface. Image originally shown in lab manual [1].

Another factor which contributes to the observed angular dependence of cosmic muons, albeit to a lesser extent, is the angle of incidence of the parent pion in the atmosphere. Muons continue traveling in the same direction of their parent particles due to conservation of momentum. If a pion enters the atmosphere at a low zenith angle, it reaches the denser lower atmosphere sooner, where it is more likely to interact with other particles, producing more pions. The increased number of pions allows for more opportunities for muons to be produced, although these tend to be lower energy due to the large number of interactions. Pions entering the atmosphere at larger zenith angle have fewer potential reaction partners, and so are more likely to decay directly into muons, as charged pions only have a mean lifetime of $0.028\mu s$ [1]. Thus, a fewer number of muons are produced, but they tend to be higher energy.

These two factors contribute to the rate of cosmic muons observed at the Earth's surface having the following dependence on the zenith angle [1]:

$$R_0 \propto \cos^2 \theta_{zenith} \tag{1}$$

with R_0 being the muon rate and θ_{zenith} being the zenith angle.

Muons and anti-muons decay into electrons and positrons, respectively, plus a muon neutrino and an electron antineutrino. That is:

$$\mu^{\pm} \rightarrow e^{\pm} + \bar{\nu}_e + \nu_\mu \tag{2}$$

The rate of muon decays can be described with the following equation:

$$R(t) = R_0 e^{-t/\tau} \tag{3}$$

with R_0 as described above, and τ being the mean muon lifetime, $2.2\mu s$ [1]. That is, given a large number of muon decays, the distribution as a function of individual decay times will follow a distribution described by Equation (3). Interestingly, the muon decay is one of the only processes uniquely involving the Weak force, and so serves as probe of the strength of the Weak interaction. This is characterized by the Fermi coupling constant, which is approximately [1]:

$$\frac{G_F}{(\hbar c)^3} = \sqrt{\frac{192\pi^3\hbar}{(m_\mu c^2)^5 \tau}} \tag{4}$$

with $m_\mu c^2$ being the muon rest energy, 105.7 MeV, and τ the muon lifetime as before. Thus $\frac{G_F}{(\hbar c)^3}$ is given by $1.166\,378\,7(6) \times 10^{-5}\text{GeV}^{-2}$ [1].

3 Experimental Methods

3.1 Description of apparatus

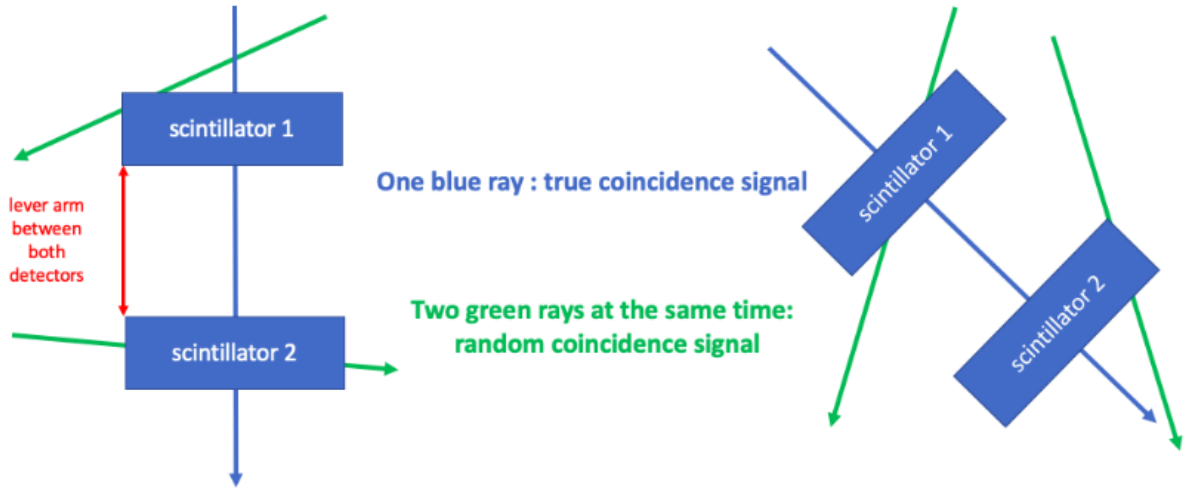


Figure 2: Diagram of the scintillator telescope. The telescope is comprised of two scintillators which can be rotated about an axis. The left side shows the telescope in the vertical position, while the right side shows it at an angle. The green arrows indicate the path of travel of random coincidence events, while the blue arrow shows the path of travel of a true coincidence event. Image originally shown in lab manual [1].

This experiment involves the use of a scintillator telescope, comprised of two scintillator paddles separated by a lever arm. Charged particles, including muons, traveling through

the scintillator material will deposit energy via the Coulomb interaction, producing a flash of light. A PMT is attached at each end of each scintillator paddle so that light pulses in the scintillator material produce an electrical signal. The apparatus can be rotated to detect particles traveling at different zenith angles by recording coincidence signals between the two paddles. However, it is possible to have two separate particles pass through each paddle at the same time, producing a coincidence signal. The number of random coincidences is independent of the zenith angle and must be subtracted from the total number of coincidences to obtain the signal counts. A diagram of the coincidence events is shown in Figure 2.

Signals from each PMT are fed to an amplifier and a discriminator. The discriminators reject signals below a threshold voltage and must be tuned so that the threshold voltage is high enough to filter out noise, but not so high that true events are rejected. After the discriminators, the signal is fed to a coincidence unit between the two PMTs on the same paddle as a noise reduction measure. From there, another coincidence unit only accepts signals coincident in both scintillator paddles. These 4-fold coincidences are the muon signals, plus the random coincidences.

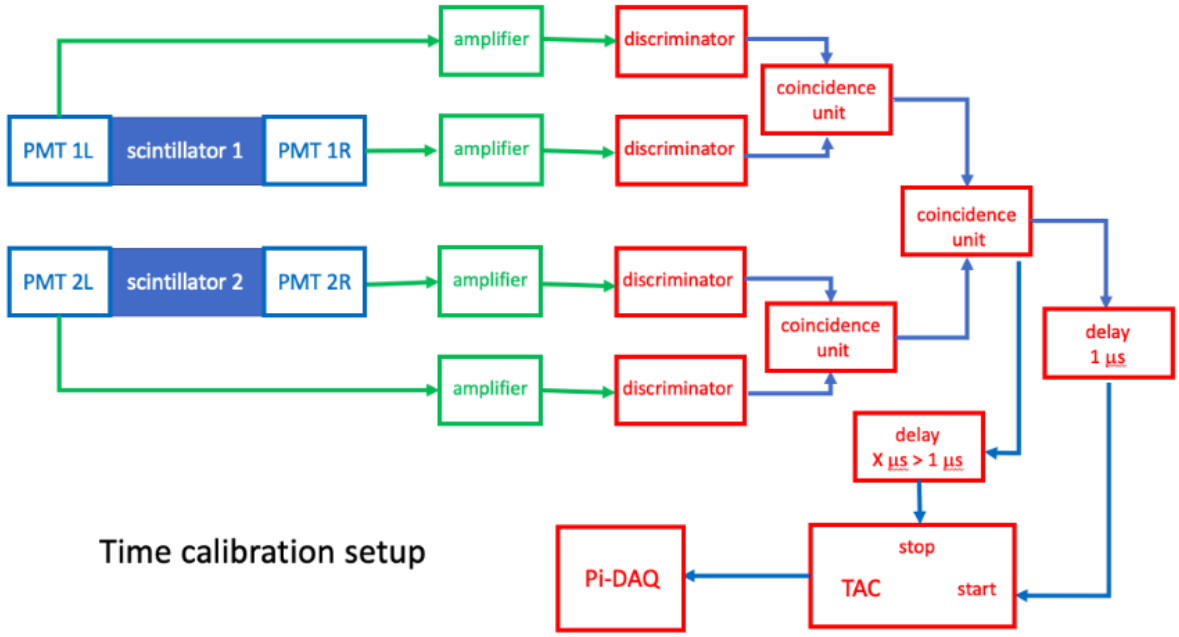


Figure 3: Block diagram of the electronics chain during the TAC calibration process. The start signal of the TAC is provided by a 4-fold coincidence from the scintillator telescope. The start signal is duplicated and has a delay added to it to produce the stop signal. Image originally shown in lab manual [1].

The 4-fold coincidence signals are then fed to a TAC module, which produces an output voltage proportional to the time measured between a start and stop signal. The TAC must be calibrated with known delay times. A block diagram of the electronics for the

TAC calibration is shown in Figure 3. If a muon decays in the scintillator material it will produce an additional light pulse which is only observed in one of the two scintillator paddles. Therefore, to measure muon decays, a 4-fold coincidence is used to detect the muon and generate the start signal for the TAC, while a 2-fold coincidence in the second (bottom) scintillator is used to detect the decay and generate the TAC stop signal. The electronics block diagram for muon decay measurement is shown in Figure 4.

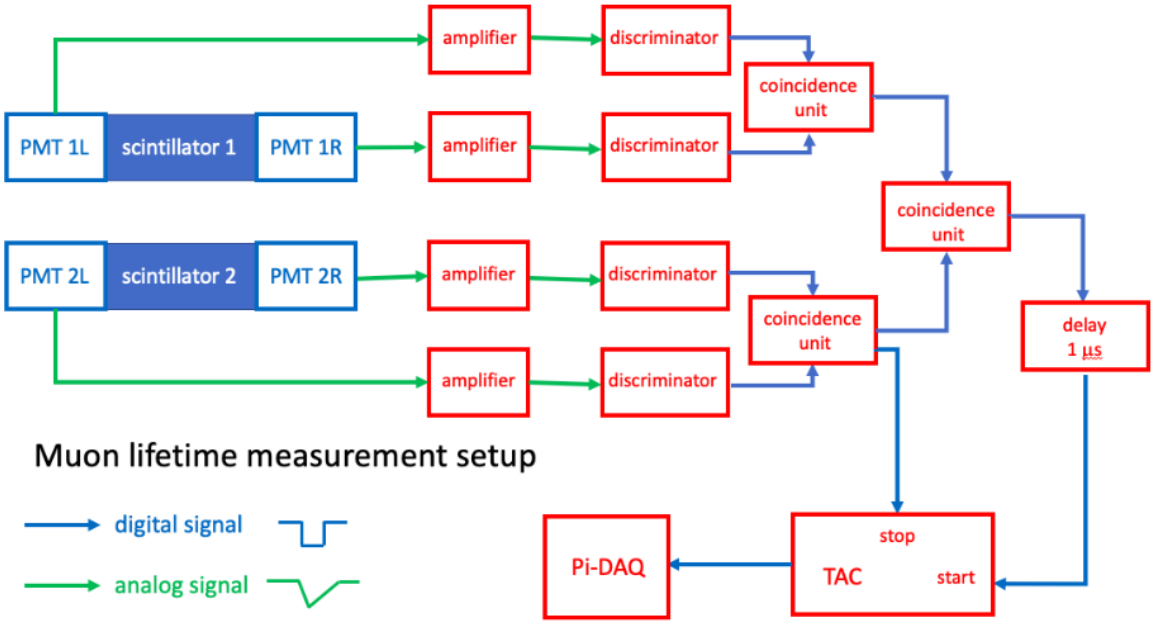


Figure 4: Block diagram of the electronics chain for measuring muon decays. The start signal of the TAC is again provided by a 4-fold coincidence from the scintillator telescope. However, the stop signal is now produced by a separate 2-fold coincidence from the second scintillator paddle. Image originally shown in lab manual [1].

3.2 Description of experimental procedure

First, the discriminator threshold voltages for each PMT must be determined. To do this, the discriminator voltages were varied from 10mV to 170mV and the number of counts over 60 seconds for the corresponding PMT was recorded. The number of counts was plotted against the threshold voltage, and the threshold voltage was chosen to be about 20mV above where the counts appeared to stabilize. The discriminator was then set to the selected value for the remainder of the experiment, and the whole process was repeated for all 4 discriminators.

Once the discriminator voltages were established, the counts were recorded for 60 seconds for each PMT individually, for 2-fold coincidences between PMTs on the same scintillator for each scintillator paddle, and for 4-fold coincidences between all PMTs. The scintillator telescope was at a zenith angle of approximately 10° for this measurement. The angular dependence of the muon rates was then observed by adjusting the angle of the scintillator

telescope from 0° to 90°, taking data for 180 seconds, and recording the number of 4-fold coincidences at each angle.

Following this, the TAC was calibrated. This was accomplished by adjusting the delay between the start and stop signals into the TAC to desired values between 1 and 8μs using an oscilloscope. 4-fold coincidences were used as the start signal and duplicated with the desired offset to produce the stop signal. The voltage output of the TAC was then recorded in a histogram using the Pi-DAQ. Data was taken for 300 seconds at each of the chosen delays for this procedure. The muon lifetime measurement was not executed because the technical limitations of the scintillator telescope make doing so take a prohibitively long time.

3.3 Description of observed phenomena

When establishing the discriminator threshold voltages, each demonstrated a clear transition from detecting noise to a plateau of relative stability with signal detection. Some of the PMTs were noisier than others, and some had relative independence of signal counts on voltage once noise was removed, while others displayed a persistent slope with increasing voltage even above noise level. The angular dependence of the muon counts was clearly demonstrated, closely following a cosine-squared relationship to the zenith angle. The TAC calibration produced a histogram with 8 sharp peaks, one for each delay setting, with no background noise.

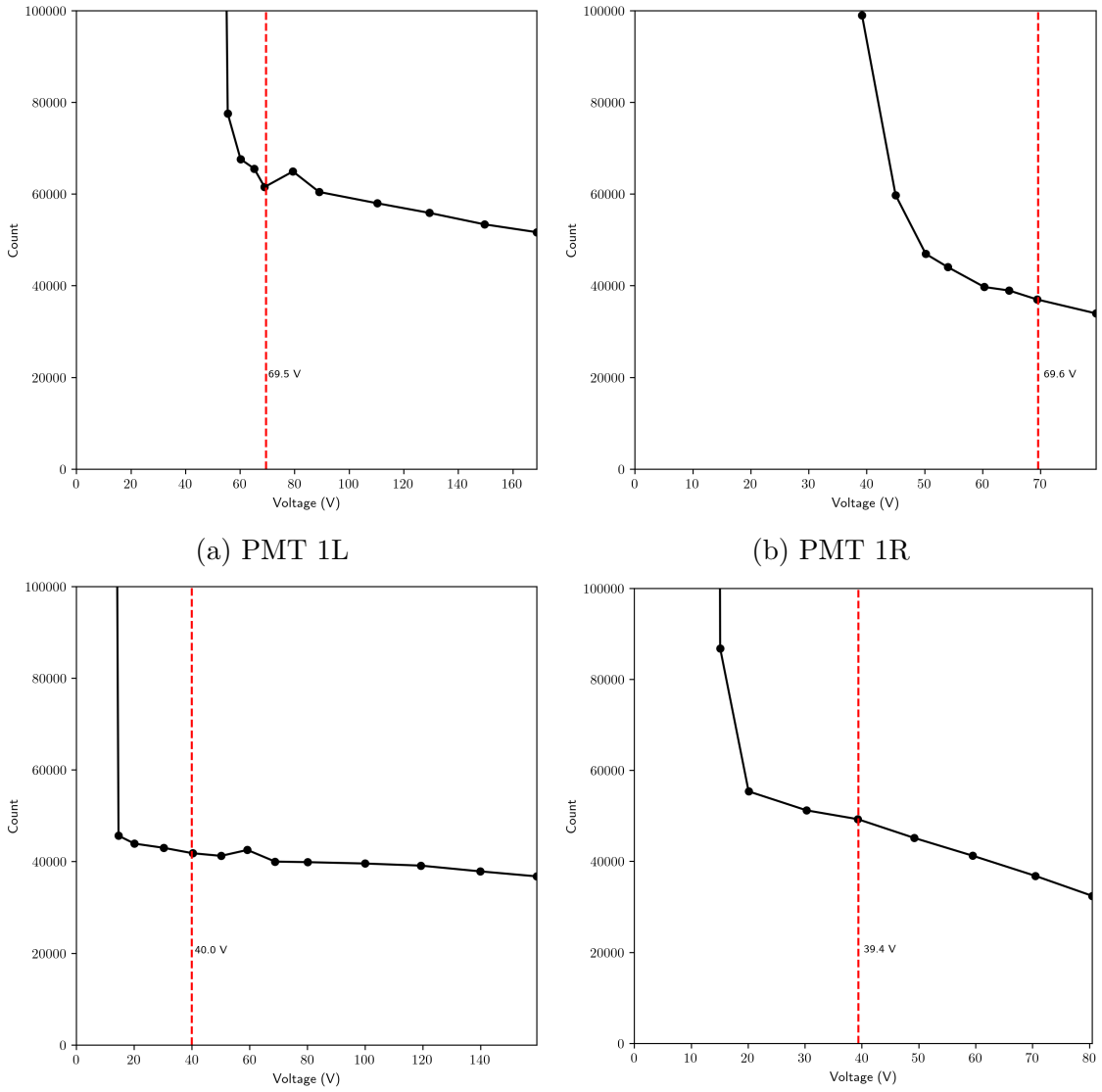
3.4 Description of sources of errors

The non-constant voltage dependence of signal rates above the noise level of some of the discriminators makes it difficult to quantify with certainty the number of true events that the corresponding PMTs record, which contributes to the error of this experiment. There is also inherent jitter in the signal of the electronics system, which makes the precision time measurements necessary for this experiment have some uncertainty. The physical apparatus of the scintillator is telescope is also somewhat unwieldy, making it sometimes difficult to set its angle accurately. The error on the telescope angle was assumed to be 1°.

4 Analysis and Results

In order to determine the appropriate values for the discriminator threshold voltages, the voltage for each discriminator was varied and the number of counts the corresponding PMT received in 60 seconds at that voltage was recorded. The number of counts was then plotted against the discriminator voltage for each PMT, shown in Figure 5. A value was selected around 20mV above the noise level, where the number of counts began to plateau. The discriminators were set to these voltages, shown in Table 1 for the rest of the experiment. The PMTs are labeled according to which scintillator paddle they are attached to, detector 1 or 2, and on which side of the scintillator they are located, either the left (L) or right (R) side. PMTs 1L and 2L both displayed a clear plateau once the noise was removed, while the number of counts for PMTs 1R and 2R continued to decrease as the threshold voltage was

increased, albeit slowly. The noise was persistent in PMTs 1L and 1R to a higher voltage, around 50mV, than for PMTs 2L and 2R, where the noise was only present to about 20mV.



(a) PMT 1L

(b) PMT 1R

(c) PMT 2L

(d) PMT 2R

Figure 5: Number of counts plotted against discriminator threshold voltage. The vertical red lines indicate the voltage that was chosen to be used for the rest of the experiment.

Once the discriminator thresholds were set, the nominal counting rates for the PMTs needed to be measured, as well as coincidence rates for scintillators 1 and 2 independently, and 4-fold coincidences of both PMTs from both detectors. The counts and rates for these settings are displayed in Table 2. PMT 1L had a noticeably higher number of events than the other 3, but even at the 2-fold coincidence level of the individual detectors, the rates were much more comparable to one another. Each increase in the level of coincidence also shows a substantial reduction in rates, as is expected.

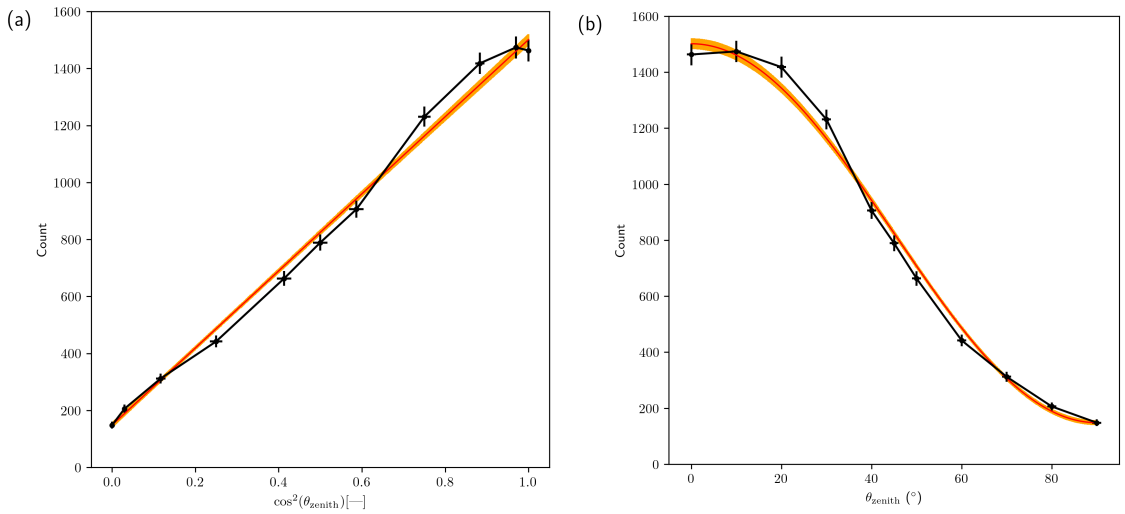
Figure 6 displays the results of the investigation of the angular dependence of muon rates.

PMT	Set discriminator voltage (mV)
1L	69.5
1R	69.6
2L	40.0
2R	39.4

Table 1: Discriminator threshold setpoints used during the experiment

Corresponding PMTs	Total counts
1L	107235
1R	32961
2L	32978
2R	46388
Detector 1 (1L && 1R)	6086
Detector 2 (2L && 2R)	7538
Detector 1 && Detector 2	296

Table 2: Counts of individual PMTs, 2-fold detector coincidences, and 4-fold coincidences in 60 seconds.



(a) Muon counts against $\cos^2\theta_{zenith}$

(b) Muon counts against θ_{zenith}

Figure 6: Angular dependence of muon counts. Figure 6a shows the counts against the cosine squared of the zenith angle, while 6b shows the counts plotted directly against the zenith angle. A linear fit is applied to the data in Figure 6a, shown by a red line, with the error on the fit shown by the yellow band. The same fit and error are propagated to the plot in Figure 6b.

In Figure 6a, the number of events recorded at each angle are plotted against $\cos^2\theta_{zenith}$,

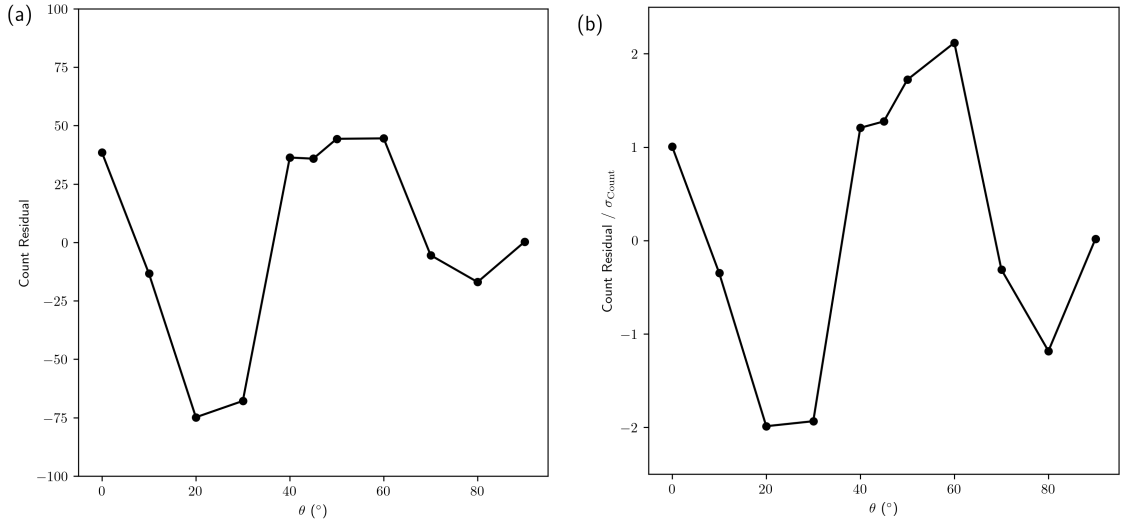
with a linear fit applied to the data and the error on the fit shown as well. Figure 6b shows the same data, but with counts plotted directly against θ_{zenith}. The linear fit and error function from Figure 6a are propagated onto this plot. The fit and error functions obtained are

$$R_0(x) = (1353 \pm 23)x + (148.2 \pm 9.0), \tag{5}$$

and

$$\sigma^2_{R_0}(x) = 531.7x^2 - 235.4x + 80.81 \tag{6}$$

respectively, with $x = \cos^2 \theta_{zenith}$. The residuals of the number of counts to the fit are plotted in Figure 7a with the residuals scaled by the error shown in Figure 7b. These results show that the number of muons recorded closely follows the anticipated cosine-squared relationship with the zenith angle.



(a) Fit residuals

(b) Fit residuals divided by the error

Figure 7: Residuals of the data against the fit function. Figure 7a shows the raw residuals, while 7b shows the residuals divided by the error on the points.

Figure 8 displays the histogram used in calibrating the TAC. Each peak in the histogram corresponds to a different set value for the delay between the TAC start and stop signals. The peaks were analyzed to obtain the mean channel number and standard deviation for each in order to perform the calibration. The peaks are quite clear with no measured background, although as the delay was increased, the spread of the peaks increased as well. Figure 9a shows the calibration of the TAC, with the set delay plotted against the channel number of each peak and a linear fit applied to obtain the calibration function. Figure 9b shows the residuals of the calibration. The calibration function and error on the calibration are, respectively

$$t = (0.001027 \pm 5.9 \times 10^{-6})x + (0.020 \pm 0.026) \tag{7}$$

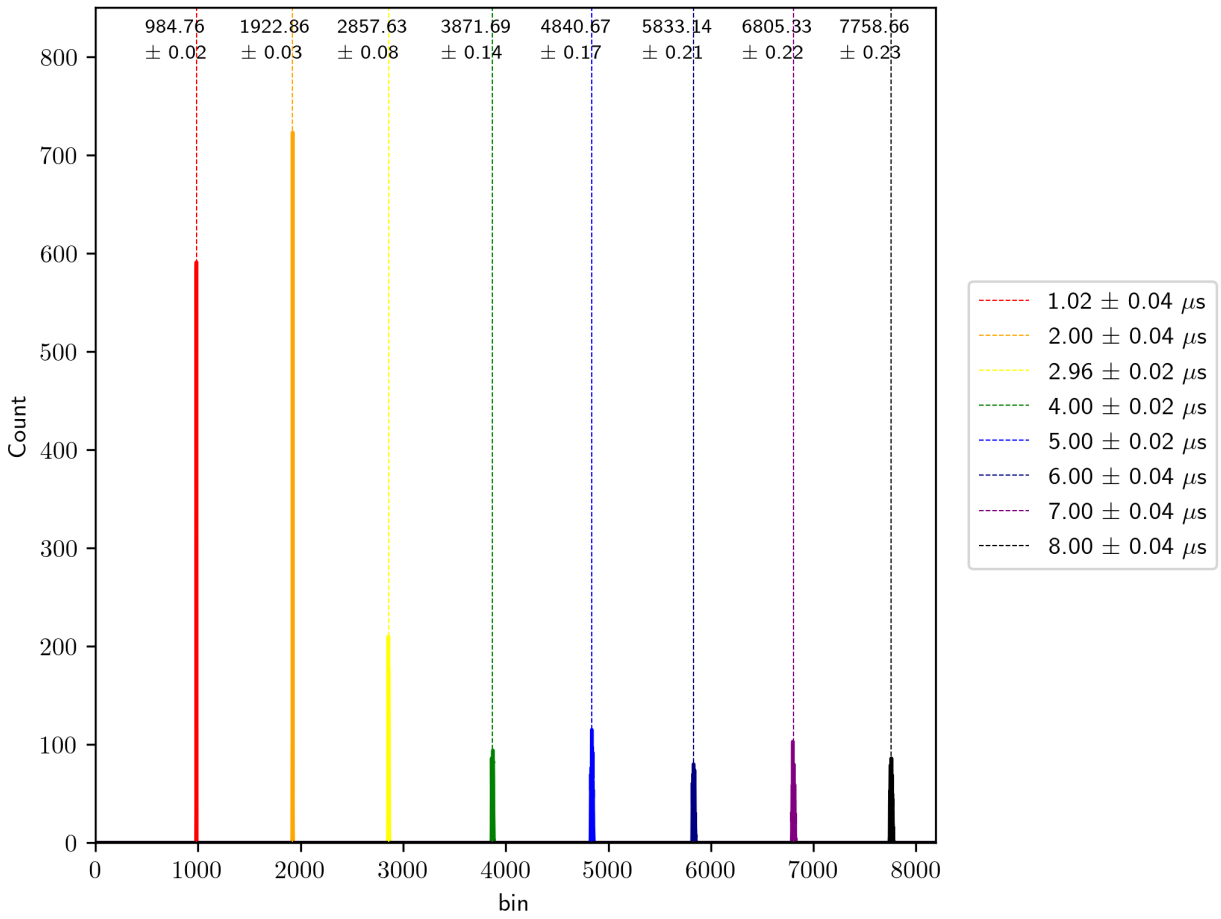
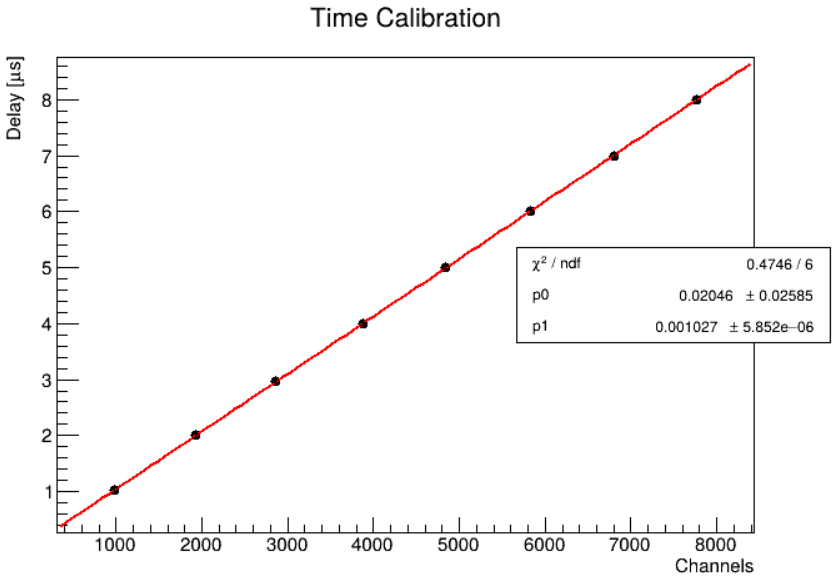


Figure 8: Histogram of the TAC calibration data. Each delay corresponds to a different peak. Mean bin values and standard deviations of each peak are shown at the top of the plot, while the set delay times and errors corresponding to each peak are shown on the right.

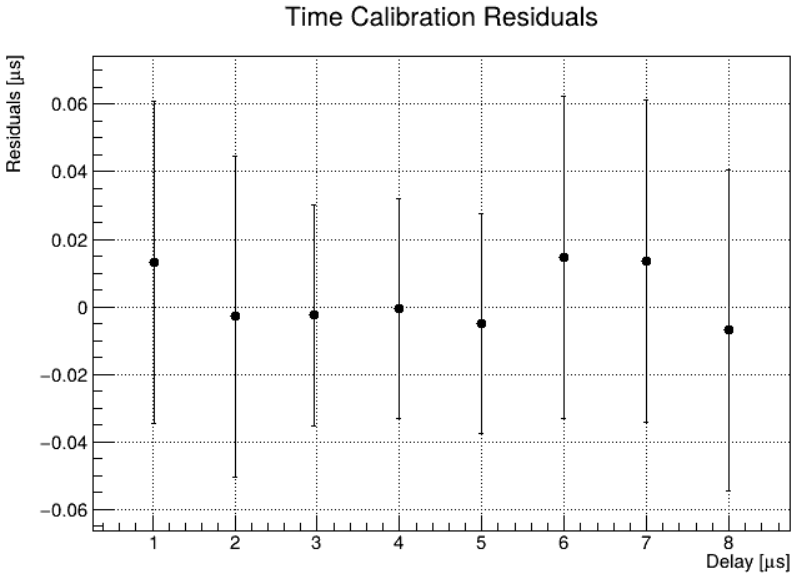
and

$$\sigma_t^2 = 3.424 \times 10^{-11} x^2 - 2.804 \times 10^{-7} + 6.678 \times 10^{-4}, \tag{8}$$

with x being TAC channels. The residuals of the calibration are quite small compared to the estimated errors, indicating that the precision on the calibration is potentially higher than the error bars may indicate. This calibration would be a good starting point to measure the muon lifetime given the opportunity.



(a) Calibration fit



(b) Calibration residuals

Figure 9: TAC calibration results and residuals.

5 Conclusions

In this experiment, the angular dependence of cosmic muon rates was investigated. This was accomplished with the use of a scintillator telescope with proper setup and calibration. First, appropriate discriminator thresholds were determined by scanning through the voltages on the discriminators for each PMT and recording the number of counts received. Then, nominal rates were measured for each PMT individually, the 2-fold coincidence of PMTs on each scintillator paddle, and the 4-fold coincidence of all PMTs. After that, the rate of cosmic muons was measured at various zenith angles and fitted to the cosine squared of the zenith angle with good agreement. Finally, the TAC was calibrated to a high degree of precision. This last portion would be an important step in preparation for measuring the rate of muon decays. However, the current setup of the scintillator telescope makes measuring muon decays difficult without taking data for a prohibitively long period of time. It could be worthwhile to attempt the muon decay measurement, provided fewer time constraints, or with a modified telescope with more scintillating material to increase the odds of capturing muon decays in the detector.

References

[1] P.M. King G. Jensen and J. Roche. *PHYS6751: Nuclear Graduate Lab Lab manual v24.5*. Physics and Astronomy Department, Ohio University, 2024, pp. 37–46.